Unit 4 – Blockchain and Decentralized Applications

Lab 2 Manual

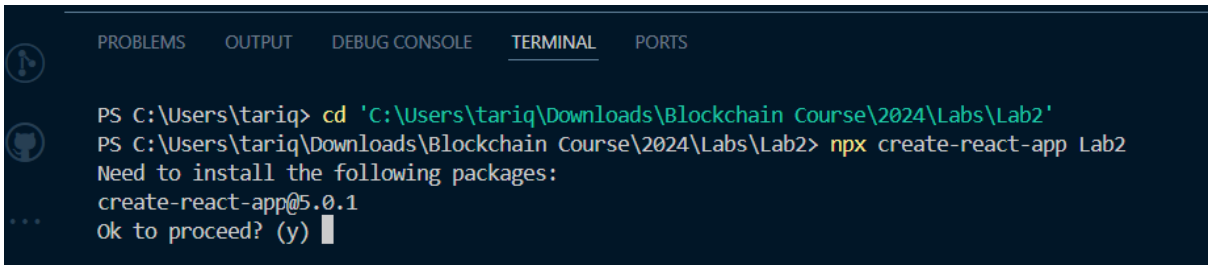
Introduction to Solidity Programming with VS Code and Truffle

Objective:
In this lab, students will set up a local blockchain environment using Truffle and Ganache, deploy a simple smart contract, and interact with it through the Truffle console.
By the end of this lab, students will:

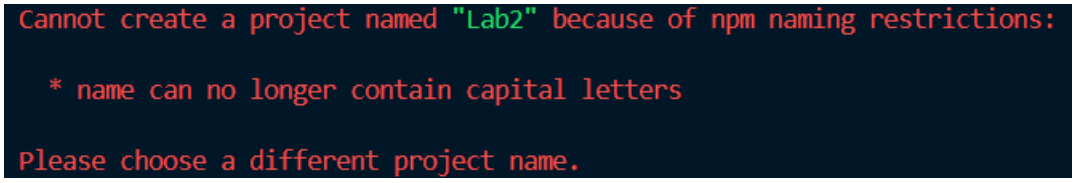
- Understand the basics of Solidity using VS Code.
- Write a simple smart contract and interact with it.
- Deploy and interact with the contract using VS Code and Ganache blockchain.

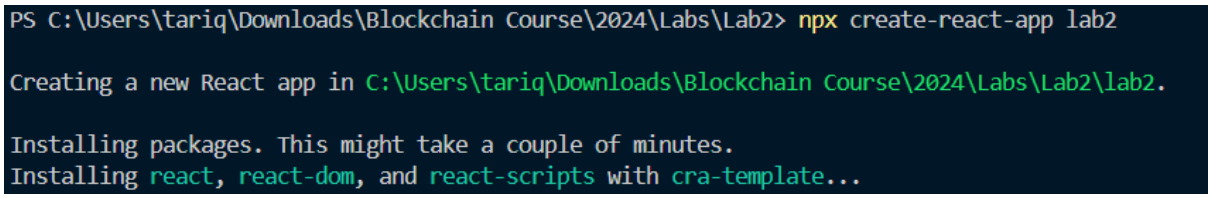
Step 1: Setting Up VS Code Environment

- Click on View menu, select Terminal from the drop down menu.
- Type *mkdir Lab2* by creating a folder
- Type *cd Lab2* to get into the directory
- Then, type *npm -v* to find out if you have npm installed if yes then type *npx create-react-app Lab2* to create a react application



- If the name of the project is in capital letters, I will throw an error, so always use the name in lowercase letter.





- It asks for confirmation, press ‘y’.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.

Created git commit.

Success! Created lab2 at C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2
Inside that directory, you can run several commands:

  npm start
    Starts the development server.

  npm run build
    Bundles the app into static files for production.

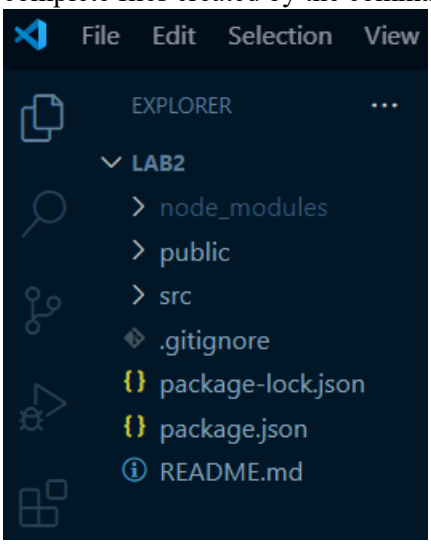
  npm test
    Starts the test runner.

  npm run eject
    Removes this tool and copies build dependencies, configuration files
    and scripts into the app directory. If you do this, you can't go back!

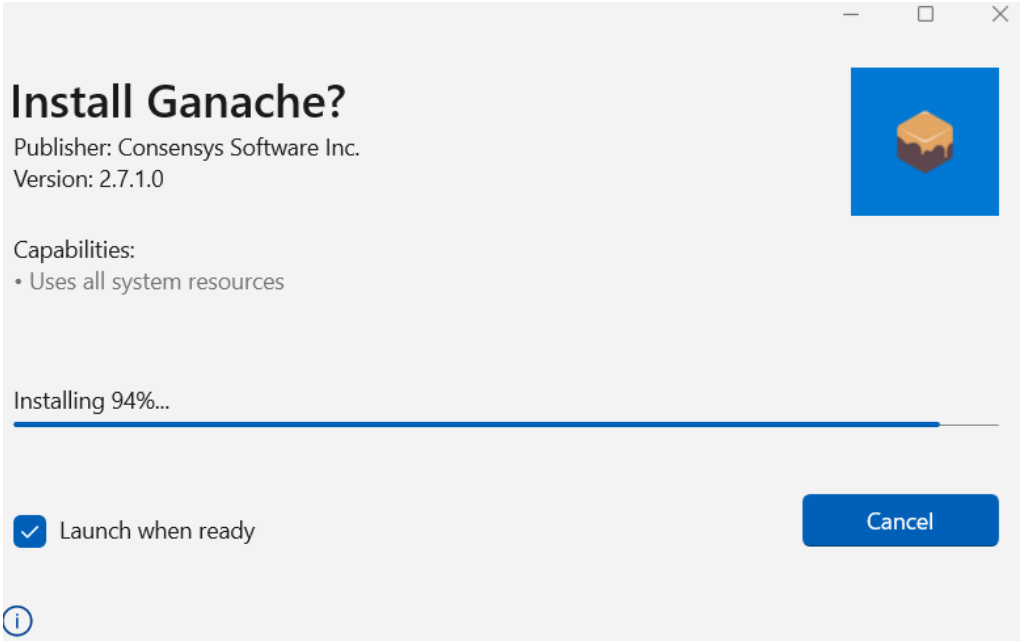
We suggest that you begin by typing:

  cd lab2
  npm start

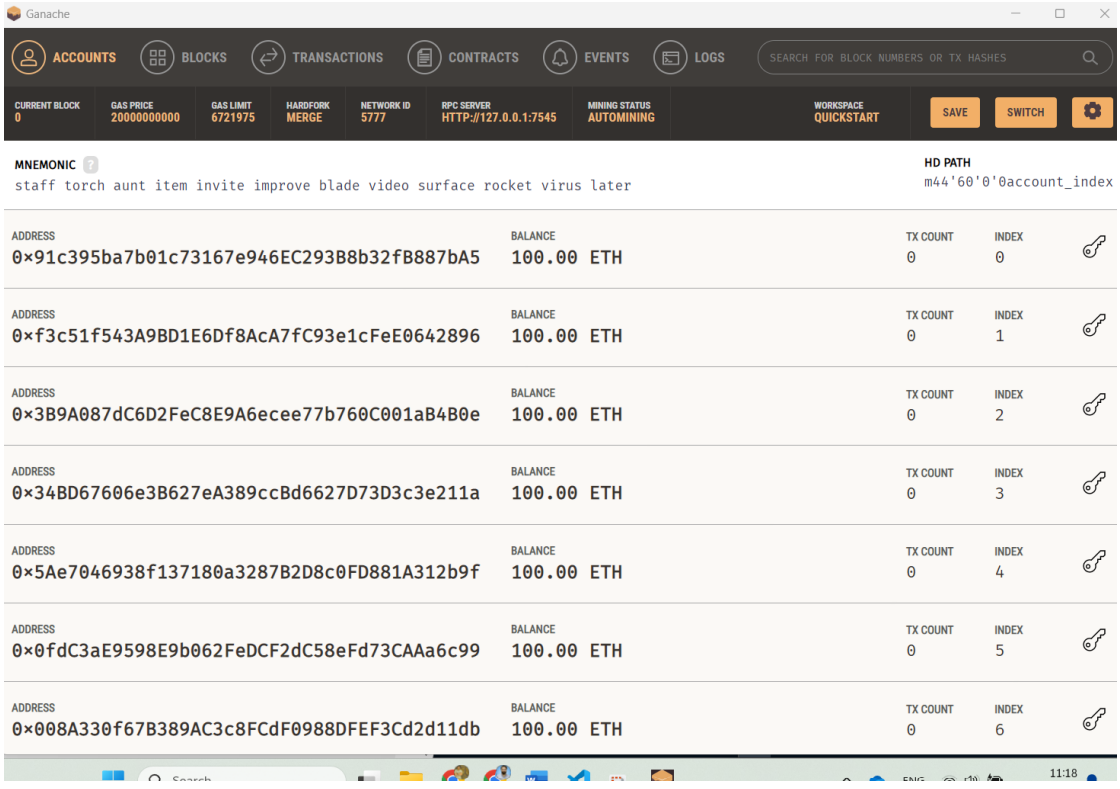
Happy hacking!
PS C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2>
```

- After the installation is complete, close the terminal.
- Go to File menu and click on Open Folder. You can go to the directory. You will see the complete files created by the command in explorer area.
- 

Step 2: Install Ganache from [Ganache]



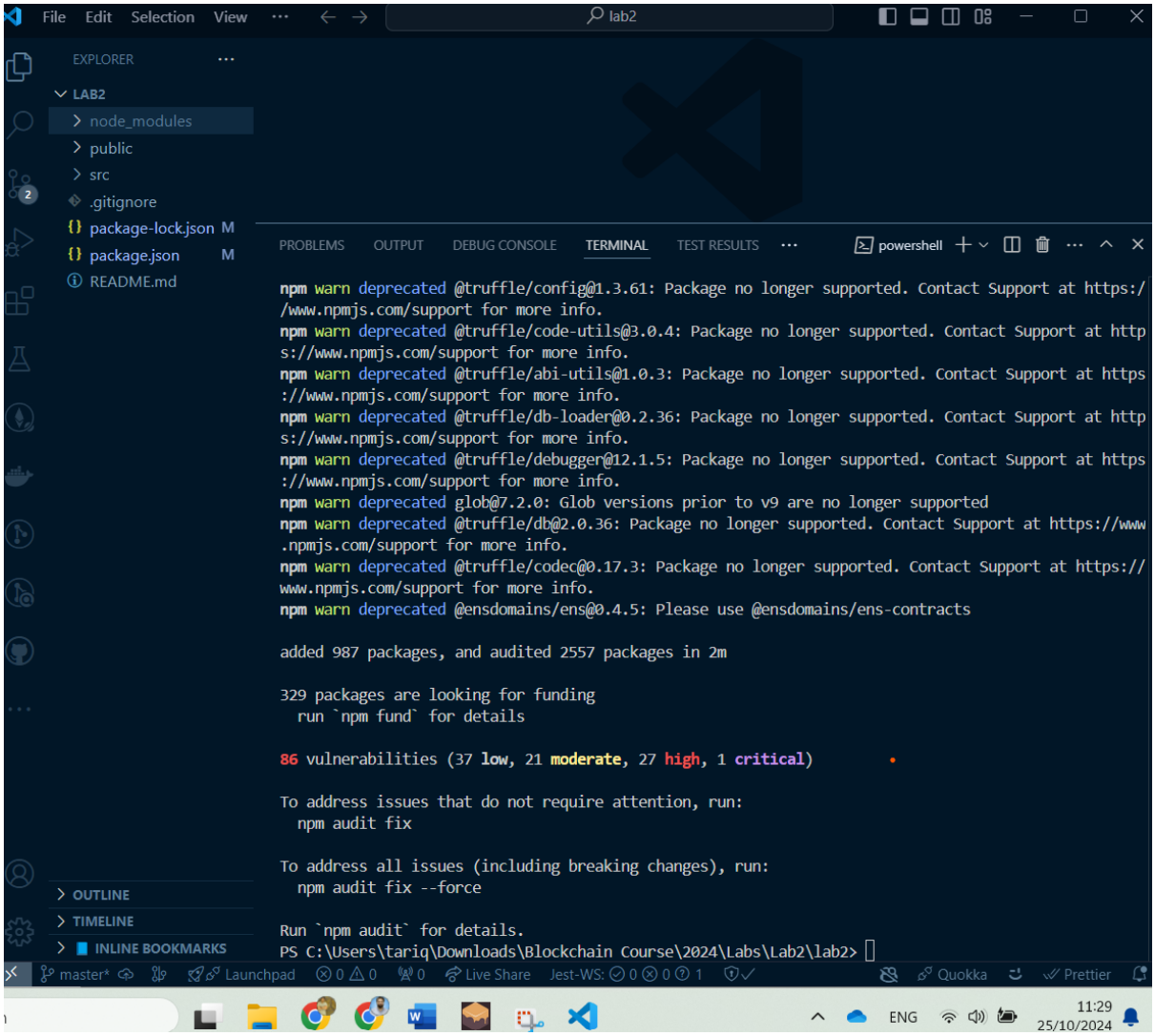
After installing, click on quick start and you will see the interface somewhat like this:



please do not close this application. We want it running at the backend.

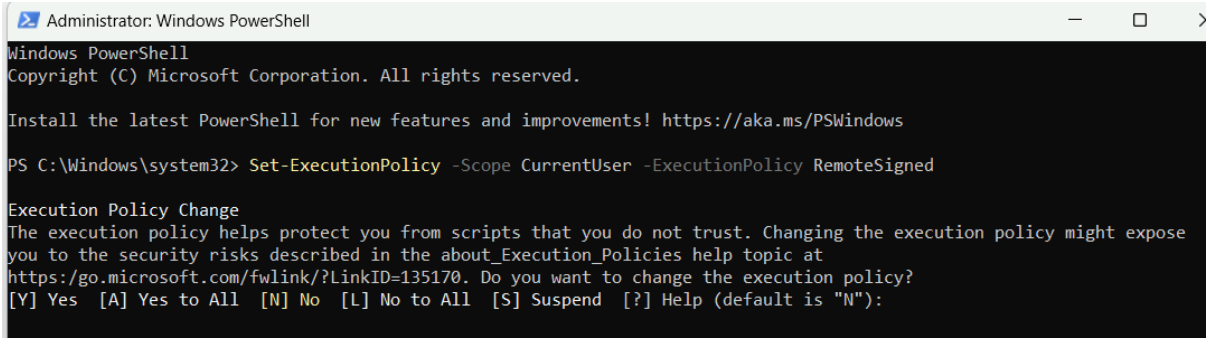
Step 3: Come back to your VS Code terminal

Type *npm install truffle*



After the installation is complete, please close the terminal and open a new terminal.

Step 4:
Type *npm install -g truffle* to install truffle in your environment. Sometimes it gives issues so, please permit your running scripts enabled on your system.
Set the execution policy:
set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned



Revert the execution policy:
set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy Restricted

The truffle will be installed successfully

```
PS C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2> npm install -g truffle
npm warn deprecated testrpc@0.0.1: testrpc has been renamed to ganache-cli, please use this pack
ge from now on.
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use i
. Check out lru-cache if you want a good and tested way to coalesce async requests by a key valu
, which is much more comprehensive and powerful.
npm warn deprecated @truffle/source-map-utils@1.3.119: Package no longer supported. Contact Supp
rt at https://www.npmjs.com/support for more info.
npm warn deprecated mkdirp-promise@5.0.1: This package is broken and no longer maintained. 'mkdi
p' itself supports promises now, please switch to that.
npm warn deprecated rimraf@2.7.1: Rimraf versions prior to v4 are no longer supported
npm warn deprecated @truffle/db-loader@0.2.36: Package no longer supported. Contact Support at h
tps://www.npmjs.com/support for more info.
npm warn deprecated @truffle/promise-tracker@0.1.7: Package no longer supported. Contact Support
at https://www.npmjs.com/support for more info.
npm warn deprecated har-validator@5.1.5: this library is no longer supported
npm warn deprecated @truffle/error@0.2.2: Package no longer supported. Contact Support at https:
/www.npmjs.com/support for more info.
npm warn deprecated glob@7.2.0: Glob versions prior to v9 are no longer supported
npm warn deprecated apollo-datasource@3.3.2: The `apollo-datasource` package is part of Apollo S
rver v2 and v3, which are now deprecated (end-of-life October 22nd 2023 and October 22nd 2024, r
spectively). See https://www.apollographql.com/docs/apollo-server/previous-versions/ for more de
ails.
changed 1160 packages in 48s

95 packages are looking for funding
  run `npm fund` for details
PS C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2> truffle version
Truffle v5.11.5 (core: 5.11.5)
Ganache v7.9.1
Solidity v0.5.16 (solc-js)
Node v20.18.0
Web3.js v1.10.0
```

Now,
Initialize truffle by typing *truffle init*

```
PS C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2> truffle init

Starting init...
================

> Copying project files to C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2

Init successful, sweet!

Try our scaffold commands to get started:
  $ truffle create contract YourContractName # scaffold a contract
  $ truffle create test YourTestName         # scaffold a test

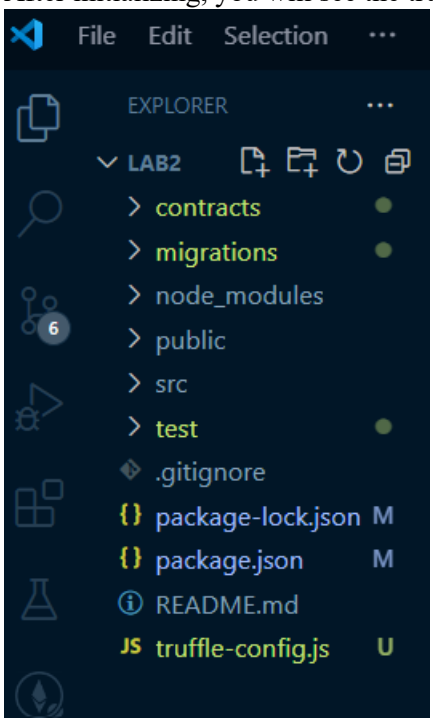
http://trufflesuite.com/docs

PS C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2>
```

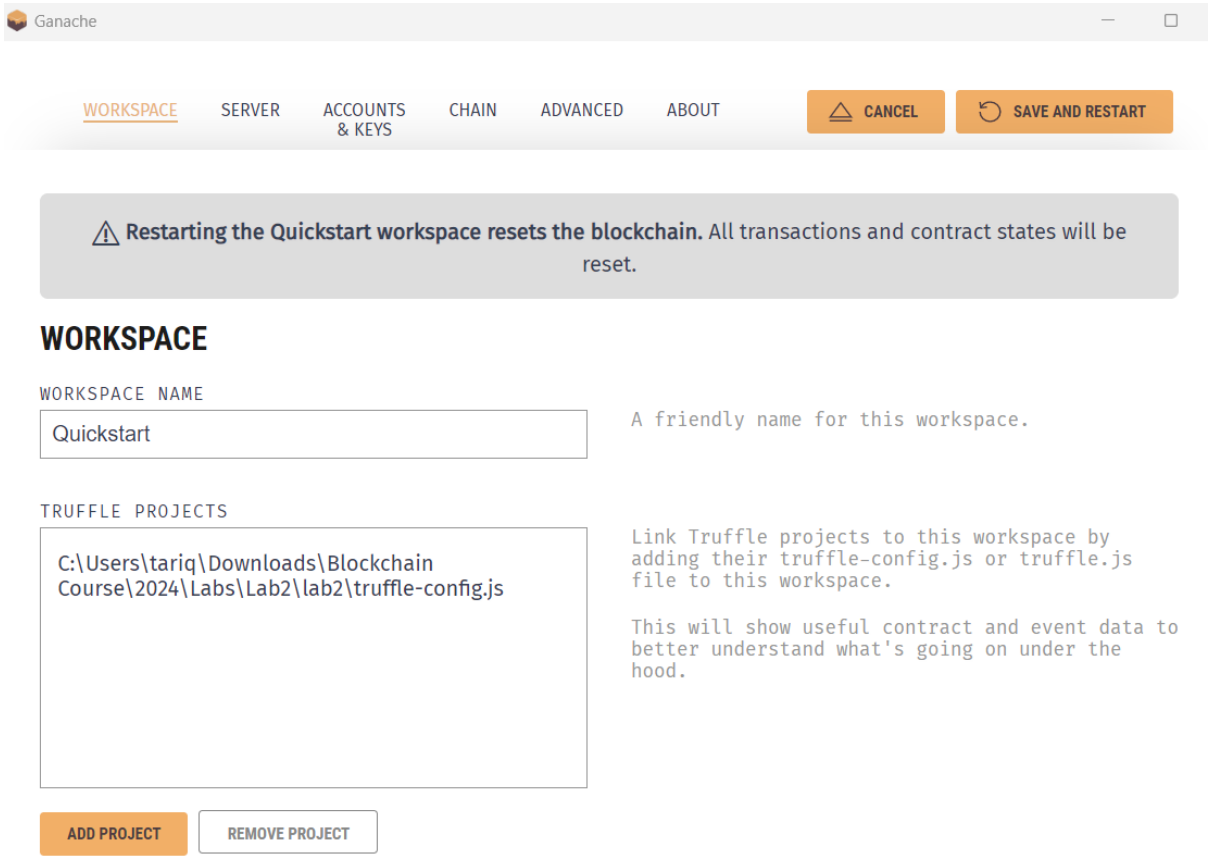
Note:
If the truffle was initialized earlier on same contract, your command will look like this:

```
PS C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2> truffle init
contracts already exists in this directory...
? Overwrite contracts? (y/N)
Starting init...
================
? Overwrite contracts? Yes
migrations already exists in this directory...
? Overwrite migrations? Yes
test already exists in this directory...
? Overwrite test? Yes
truffle-config.js already exists in this directory...
? Overwrite truffle-config.js? Yes
```

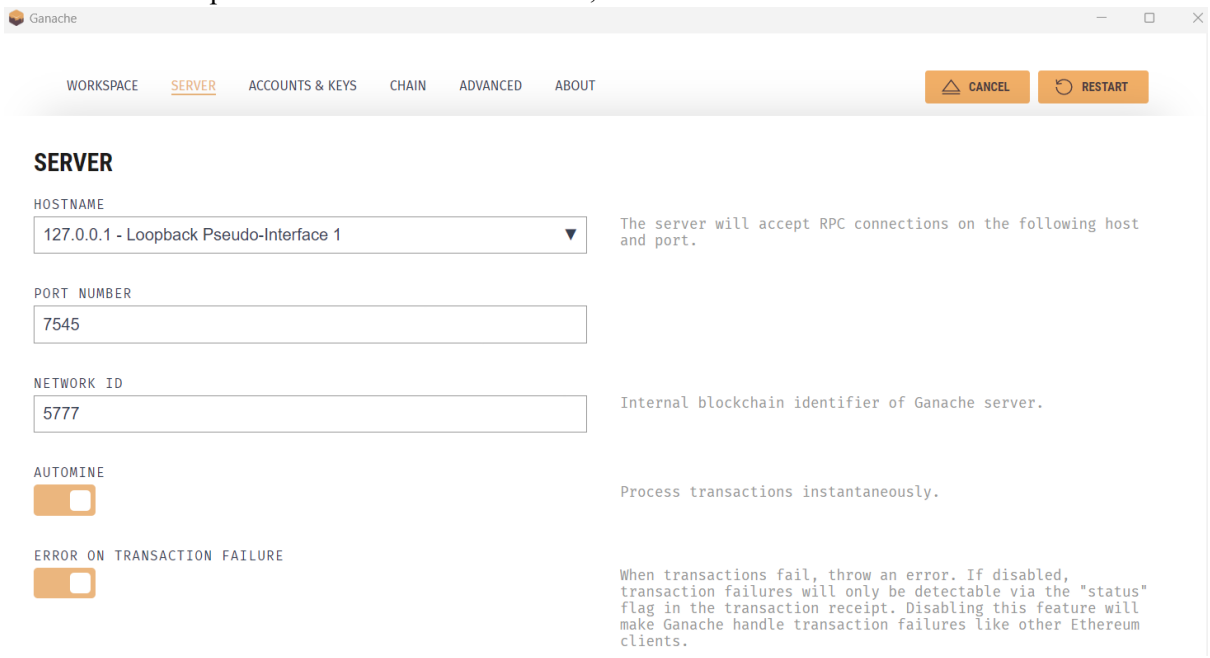
After initializing, you will see the truffle-config.js file in the explorer tab.



Step 5: Open Ganache, go to settings and add your project to it by adding the truffle-config.js file.



Also confirm the port number and network ID i.e.,



Then, click on save and restart.

Go back to VS Code terminal and type *truffle console*
You will see that the truffle console is open with ganache

```
PS C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2> truffle console
truffle(ganache)>
```

Add a file to your contract and save it:



Or if want, you can add the contract which we have compiled and executed in our first lab.

Also,
Add the file 1_migtestcontract.js and save it:



In the truffle console now you can type *compile*

```
PROBLEMS  OUTPUT  TERMINAL  ···                                   node

truffle(ganache)> compile

Compiling your contracts...

==========================

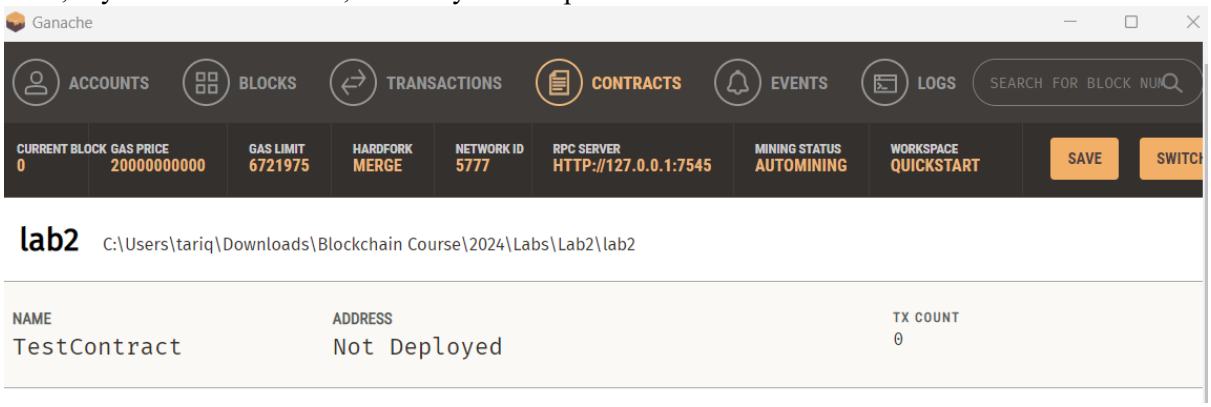
> Compiling .\contracts\testcontract.sol

> Artifacts written to C:\Users\tariq\Downloads\Blockchain Course\2024\La
bs\Lab2\lab2\build\contracts

> Compiled successfully using:
   - solc: 0.8.21+commit.d9974bed.Emscripten.clang

truffle(ganache)>
```

Now, if you see the Ganache, it holds your compiled contract



Since our contract is compiled successfully, now type *.exit* to exit from the truffle console environment.

Soon as you exit, then type *truffle migrate -reset*

```
truffle(ganache)> .exit
PS C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2> truffle migrate -reset

Compiling your contracts...
==========================
> Compiling .\contracts\testcontract.sol
> Artifacts written to C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2\build\con
tracts
> Compiled successfully using:
   - solc: 0.8.21+commit.d9974bed.Emscripten.clang
Network up to date.
PS C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2>
```

Soon as you run the command truffle migrate -reset, your contract will be deployed.

Brava! Your contract is deployed successfully. This is how it will look like in terminal:

```
PS C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2> truffle migrate -reset

Compiling your contracts...
===========================
> Compiling .\contracts\testCon.sol
> Artifacts written to C:\Users\tariq\Downloads\Blockchain Course\2024\Labs\Lab2\lab2\build\contracts
> Compiled successfully using:
   - solc: 0.8.0+commit.c7dfd78e.Emscripten.clang


Starting migrations...
======================
> Network name:    'development'
> Network id:      5777
> Block gas limit: 6721975 (0x6691b7)
```

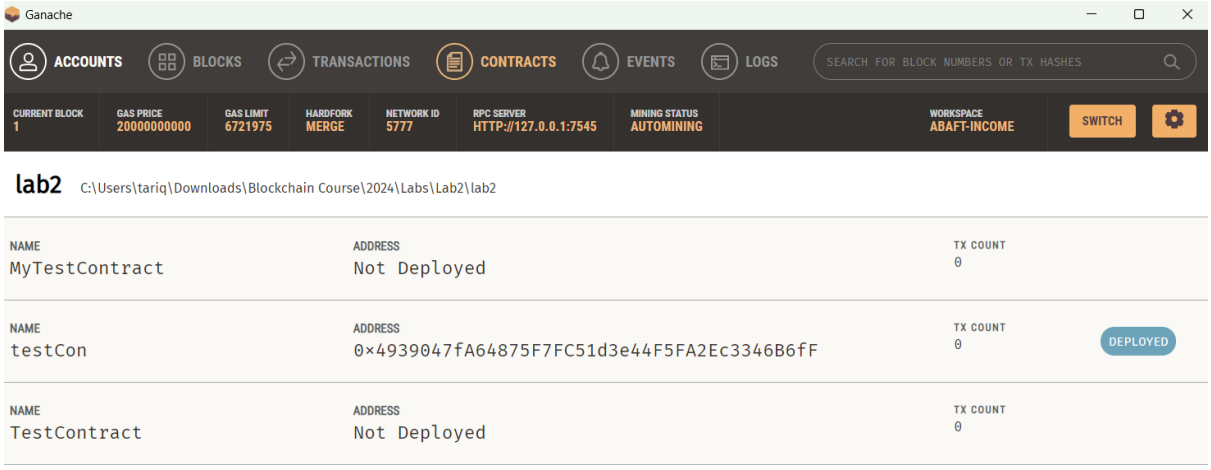
```
1_migtestCon.js
===============

   Deploying 'testCon'
   -------------------
   > transaction hash:    0x0fd08dfc396cbf7ac2a088d65710c58cb0e306de43ea966e992dc9c25719e9e0
   > Blocks: 0            Seconds: 0
   > contract address:    0x4939047fA64875F7FC51d3e44F5FA2Ec3346B6fF
   > block number:        1
   > block timestamp:     1730404782
   > account:             0xf5A160C2a83C3394E51d06f6Aab4f65dE28d778C
   > balance:             99.999473901625
   > gas used:            155881 (0x260e9)
   > gas price:           3.375 gwei
   > value sent:          0 ETH
   > total cost:          0.000526098375 ETH

   > Saving artifacts
   -------------------------------------
   > Total cost:      0.000526098375 ETH

Summary
=======
> Total deployments:   1
> Final cost:          0.000526098375 ETH
```

And, you can also see it on Ganache interface i.e.,



Step 6 (Optional):
To verify get again into the *truffle console* by typing it in the terminal
In the console, type *const acc=await TestContract.deployed()* OR
const instance = await TestContract.deployed();

Now your contract is deployed successfully and verified.
Task:
"🚀 Challenge Lab: Decentralized Dev in Action! 🚀
This week, you're invited to dive into Truffle and Ganache on VS Code—getting hands-on with core blockchain tools! Here's the URL you'll need to get started: https://ferdyhape.medium.com/remix-ide-and-ganache-a-beginners-guide-to-smart-contract-deployment-b0df68c48ae6

While this isn't a graded task, those who complete it will be rewarded with chocolate 🍫 as a small bonus. Plus, who knows? If the professor is impressed, it *might* just turn into a credited assignment!
So, take this opportunity to explore freely, experiment with settings, and start building like a true blockchain developer. We'll meet up later to chat about your findings and any new insights. I can't wait to see where you take it—have fun, and happy coding!"